

Sequences, Series, and the Binomial Theorem

Answer Key

Algebra 2

Quick Answers

- | | |
|-------------------------------------|-----------------------------------|
| 1. 35 | 7. 135 |
| 2. 36 | 8. -9, 10 |
| 3. $x^3 + 9x^2 + 27x + 27$ | 9. 243 |
| 4. 640 | 10. $a^3 + 6a^2b + 12ab^2 + 8b^3$ |
| 5. 64 | 11. 80 |
| 6. $16x^4 - 32x^3 + 24x^2 - 8x + 1$ | 12. 1 |
-

Curriculum

Level: Algebra 2

HSA-APR.C.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

- 2. *If they got 72: counted ordered pairs, computing the permutation instead of the combination*
 - 4. *If they got 10: reported the binomial coefficient alone and never raised the constant to its power*
 - 5. *If they got 12: doubled the row number instead of raising two to it*
 - 7. *If they got 45: left the constant unraised, multiplying the coefficient by three instead of by three squared*
 - 9. *If they got 32: used the plain row sum for a binomial whose second term is not one*
 - 11. *If they got 20: kept the two inside the first term unraised, so only the coefficient was used*
 - 12. *If they got 625: ignored the minus sign and added the two constants before raising*
-

1. Evaluate $\binom{7}{3}$.

Step 1 (choose the tool). A binomial coefficient counts unordered selections, so use $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ with $n = 7$, $r = 3$. **Step 2 (cancel first).** $\frac{7!}{3!4!}$: cancel the 4! against the tail of 7!, leaving $\frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1}$. **Step 3 (multiply).** $\frac{210}{6} = \boxed{35}$. Cancelling before multiplying keeps the arithmetic small enough to do without a calculator.

2. Evaluate $\binom{9}{2}$; a classmate answers 72.

Step 1 (name the classmate's count). 72 is the number of *ordered* pairs that can be drawn from 9 objects — the permutation count, in which choosing A then B is different from choosing B then A. **Step 2 (correct the count).** A binomial coefficient counts unordered selections, so every pair has been counted twice; divide by $2! = 2$. **Step 3 (compute).**

$$\binom{9}{2} = \frac{9!}{2!7!} = \frac{9 \cdot 8}{2} = \boxed{36}.$$

3. Expand $(x + 3)^3$.

Step 1 (set up the row). With $n = 3$ the coefficients are 1, 3, 3, 1, and the terms run x^3 , x^2 , x^1 , x^0 with the matching powers of 3. **Step 2 (build each term).** $1 \cdot x^3$, $3 \cdot x^2 \cdot 3$, $3 \cdot x \cdot 3^2$, $1 \cdot 3^3$. **Step 3 (collect).**

$$\boxed{x^3 + 9x^2 + 27x + 27}$$

4. Coefficient of x^2 in $(x + 4)^5$.

Step 1 (find r). The general term is $\binom{5}{r}x^{5-r}4^r$; the power of x is $5 - r$, so x^2 needs $r = 3$.

Step 2 (evaluate both pieces). $\binom{5}{3} = 10$ and $4^3 = 64$ — the constant's power is what most answers leave out. **Step 3 (multiply).** $10 \cdot 64 = \boxed{640}$. Reporting 10 alone would give the count of terms of that shape, not the coefficient.

5. Sum of row 6 of Pascal's triangle.

Step 1 (write the row as a sum). The row is $\sum_{r=0}^6 \binom{6}{r}$. **Step 2 (use the theorem).** Substituting $p = 1$ and $q = 1$ into $(p + q)^6 = \sum \binom{6}{r}p^{6-r}q^r$ turns every power into 1, so the expansion collapses to exactly that sum of coefficients, and the left side becomes 2^6 . **Step 3 (compute).** $2^6 = \boxed{64}$; adding the entries $1 + 6 + 15 + 20 + 15 + 6 + 1$ gives 64 as a check.

6. Expand $(2x - 1)^4$.

Step 1 (identify p and q). Here $p = 2x$ and $q = -1$, so both the 2 and the minus sign travel with their factor through every term. **Step 2 (build the terms).** Coefficients 1, 4, 6, 4, 1 multiply $(2x)^4$, $(2x)^3(-1)$, $(2x)^2(-1)^2$, $(2x)(-1)^3$, $(-1)^4$, giving $16x^4$, $-32x^3$, $24x^2$, $-8x$, 1. **Step 3 (collect).**

$$\boxed{16x^4 - 32x^3 + 24x^2 - 8x + 1}$$

The signs alternate because q is negative and its power climbs by one each term.

7. Coefficient of y^4 in $(3 + y)^6$.

Step 1 (find r). With $p = 3$ and $q = y$, the general term is $\binom{6}{r}3^{6-r}y^r$, so y^4 needs $r = 4$.

Step 2 (evaluate both pieces). $\binom{6}{4} = 15$, and the exponent on 3 is $6 - 4 = 2$, so the constant factor is $3^2 = 9$. **Step 3 (multiply).** $15 \cdot 9 = \boxed{135}$. The exponent on 3 comes from the *other* factor of the term: the two exponents must always add to 6.

8. Solve $\frac{n(n-1)}{2} = 45$.

Step 1 (clear the fraction). $n(n-1) = 90$, so $n^2 - n - 90 = 0$. **Step 2 (factor).** $(n-10)(n+9) = 0$, giving the two solutions $\boxed{n=10}$ and $\boxed{n=-9}$. **Step 3 (interpret).** A count of teams must be a positive integer, so $n = 10$ is the answer to the tournament question; -9 solves the equation but not the situation. Check: $\binom{10}{2} = 45 \checkmark$.

9. Sum of the coefficients of $(1+2x)^5$.

Step 1 (name the substitution). The sum of the coefficients of any polynomial in x is its value at $x = 1$, because every power of 1 is 1. **Step 2 (substitute).** $(1+2 \cdot 1)^5 = 3^5$. **Step 3 (compute).** $3^5 = \boxed{243}$. The familiar 2^n row sum is only the special case where the second term is 1; here it is 2, so the base is 3.

10. Expand $(a+2b)^3$.

Step 1 (identify p and q). $p = a$ and $q = 2b$, so each term carries a power of 2 as well as a power of b . **Step 2 (build the terms).** Coefficients 1, 3, 3, 1 multiply a^3 , $a^2(2b)$, $a(2b)^2$, $(2b)^3$, giving a^3 , $6a^2b$, $12ab^2$, $8b^3$. **Step 3 (collect).**

$$\boxed{a^3 + 6a^2b + 12ab^2 + 8b^3}$$

11. Coefficient of x^3y^2 **in** $(2x+y)^5$.

Step 1 (find r). The general term is $\binom{5}{r}(2x)^{5-r}y^r$, and y^2 needs $r = 2$, which leaves $(2x)^3$.

Step 2 (evaluate both pieces). $\binom{5}{2} = 10$ and $(2x)^3 = 8x^3$, so the numeric factor is $10 \cdot 8$.

Step 3 (multiply). The coefficient is $\boxed{80}$. It is not the Pascal's-triangle entry 10, because the first term of the binomial is $2x$, not x , and that 2 is raised to the third power along with x .

12. Synthesis: sum of the coefficients of $(3x-2)^4$.

Step 1 (substitute). Evaluate the polynomial at $x = 1$: $(3 \cdot 1 - 2)^4 = (3 - 2)^4$. **Step 2 (compute).** $(3 - 2)^4 = 1^4 = \boxed{1}$. **Step 3 (interpret).** The sum of the coefficients is the value of the polynomial at $x = 1$, so this expansion is worth 1 there. The individual coefficients are large — 81, -216 , 216, -96 , 16 — but they alternate in sign, so they very nearly cancel. Ignoring the minus sign and computing $(3 + 2)^4 = 625$ answers a different question entirely: it is the sum of the absolute values of the coefficients.